Pothole Detection Using Image Processing and Data Logging

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***Abstract*—Potholes are the main cause of accidents; it is crucial to identify and categorize them using image processing techniques. Different image pre-processing and segmentation methods for pothole detection were reviewed using performance measures. In this paper image pre-processing based on the difference of Gaussian-Filtering and clustering-based image segmentation methods are implemented for better outcome. According to the findings, edge detection-based segmentation is favoured for its specificity and K-Means clustering-based segmentation was selected for its quickest computation time. The primary objective of this study is to find a superior approach that is more precise and efficient than current methods.**

Keywords—pothole; image processing; detection; identification; segmentation; edge detection; data logging

I. INTRODUCTION

A depression in a road surface, typically formed of asphalt due to traffic removing shattered bits of pavement, is called a pothole. Repair strategy depends on the form, dimensions, and depths of the pothole. These are commonly found in cities with heavy motor vehicle traffic and can be caused by poor maintenance, shifting ground or surface area, improper construction or compaction, excessive use over an issue area, adverse weather conditions, and the material make-up of the road or street.

Accurate measurements are required to assess the size and severity of potholes, but manual surveying is laborious and unreliable. As a result, methods for image processing have been used to find potholes on highways. Digital images are subjected to numerous manipulation processes, including enhancement, restoration, compression, and segmentation. It has numerous uses in industries like robotics, computer vision, and medical imaging.

The size, depth, and position of potholes affect the rehabilitation process. Potholes are a common issue on roads. Image processing methods have been used to estimate potholes accurately, which is necessary for efficient repairs.The paper discusses image pre-processing, types of filters used for edge detection ,working of canny filters followed by data logging process.

II. IMAGE PRE-PROCESSING

A. Image Resizing

The original pothole image was captured in high resolution, more data was stored. Using the original pothole image may take more computational time. So, the image has to be resized into lower resolution for faster processing and optimal segmentation. The main challenge in resizing the original image is to obtain a lower resolution image without losing its required properties and not losing its quality. So, finding the resizing resolution is the first step for achieving better and faster results in the image segmentation process. When resizing, we should not use a fixed resolution, such as [200,200], as it can change the image's aspect ratio. If the aspect ratio is changed it badly affects the results So, we have to resize without finite resizing resolution for example [NaN,200], it won't affect the aspect ratio of the image and provide a resized image with original shape properties.

B. Grayscale Conversion

Next, the RGB-image (original pothole image) must be converted into a grayscale image as part of the pre-processing process using standard techniques to restrict the images to a single plane before the image segmentation process.

C. Difference of Gaussian-Filtering (DoG)

To improve edge recognition with less noise, the DoG-Filter was used as a pre-processing filter on the original pothole picture. It determines the edges in the grayscale picture by comparing the two sigma values of two Gaussian profiles.

III. TYPES OF FILTERS FOR EDGE DETECTION

A digital picture's edges with discontinuities or sharp brightness shifts are found using the image processing technique known as edge detection. The pixels near the picture's edges (or limits) are those where the brightness of the image varies most noticeably. In image processing, picture pattern recognition, and computer vision, it is a basic step. Processing digital images with extremely high resolution is made much easier by convolution techniques.

A. Sobel filter

The Sobel filter is made up of two 3x3 kernels, one for identifying edges in the x direction and one for detecting edges in the y direction. The kernels have symmetric weights around the center pixel. Using these kernels, the filter calculates the gradient of picture intensity in the x and y directions and then combines the findings to determine the magnitude of the gradient. In computer vision applications such as object recognition and tracking, the Sobel filter is extensively utilized.

```
-1 -2 -1
 0  0  0
 1  2  1
```

The matrix for detecting vertical edges is:

```
-1  0  1
-2  0  2
-1  0  1
```

Before being applied to the image, the Sobel filter is convolved with it. The values in the filter are multiplied by the corresponding pixel values in the image, and the filter is therefore centred on each pixel in the image. The value for

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the appropriate pixel in the output image is then determined using the sum of the produced values.
The bright pixels in the grayscale picture created by the Sobel filter correspond to the edges and boundaries of the original image. The magnitude of the gradient at a given pixel is determined by taking the square root of the sum of the squares of the horizontal and vertical gradients there.

B. Prewitt filter

Similar to the Sobel filter, but with different kernel coefficients, is the Prewitt filter. Although the weights of the Prewitt filter kernels are not symmetric around the center pixel, they are similarly 3x3 in size. Although the Prewitt filter produces narrower edges than the Sobel filter, it is less susceptible to noise. Applications using computer vision also frequently use it. Similar to the Sobel filter, it is a spatial filter that emphasizes the edges and bounds of an image.
Two 3x3 matrices, one for identifying horizontal edges and one for detecting vertical edges, make up the Prewitt filter. The following matrix is used to find horizontal edges:

```
-1 -1 -1
 0  0  0
 1  1  1
```

The matrix for detecting vertical edges is:

```
-1  0  1
-1  0  1
-1  0  1
```

C. Laplacian filter

The Laplacian filter calculates the second derivative of the image intensity to find areas of rapid intensity change. The Laplacian filter can identify edges with extreme precision and in any direction, although it is more noise sensitive than the Sobel and Prewitt filters. Before using other edge detection methods, the Laplacian filter is widely used as a pre-processing step. A 3x3 matrix is frequently used to depict the Laplacian filter:

```
0  1  0
1 -4  1
0  1  0
```

By convolving it with the image, the Laplacian filter is applied to the image. The values in the filter are multiplied by the corresponding pixel values in the image, and the filter is therefore centered on each pixel in the image. The value for the appropriate pixel in the output image is then determined using the sum of the produced values.

Bright pixels in the grayscale image created by the Laplacian filter correspond to regions of rapid intensity shift in the original image. The Laplacian filter, however, is prone to noise, which might lead to spurious edges in the final image. In order to enhance the edges before using further filters, it is typically used in conjunction with other filters or as a pre-processing step. Additionally, by re-adding the filtered image to the original image, the Laplacian filter can be used to sharpen images.

D. Roberts cross filter

The Roberts cross filter is a fundamental edge detection filter that uses two 2x2 kernels to find edges that run diagonally. The kernel weights are not symmetrical about the central pixel. The filter identifies edges less precisely but is less vulnerable to noise than the Sobel and Prewitt filters. It uses a first-order spatial filter to find edges that are oriented at 45 degrees.

Two 2x2 matrices, one for detecting diagonal edges from top left to bottom right and the other for detecting diagonal edges from top right to bottom left, make up the Roberts cross filter. The following matrix is used to find diagonal edges running from top left to bottom right:

```
1  0
0 -1
```

The matrix for detecting diagonal edges from the top right to the bottom left is:

```
 0  1
-1  0
```

Convoluting the Roberts cross filter with the image allows it to be applied to the image. The values in the filter are multiplied by the corresponding pixel values in the image, and the filter is therefore centered on each pixel in the image. The value for the appropriate pixel in the output image is then determined using the sum of the produced values.

The Roberts cross filter creates a grayscale image with bright pixels aligned at a 45-degree angle to the boundaries of the source image. The gradient's strength at each pixel in the image is determined by the absolute difference between the diagonal pixels used in the filter. Although it is less efficient than more intricate edge detectors like the Sobel or Canny, the Roberts cross filter is quick and easy to use.

E. Marr-Hildreth edge detector

The Laplacian of Gaussian (LoG) filter is used by the Marr-Hildreth edge detector to detect edges at various scales. To find potential edge sites, the LoG filter is first applied to the image. Then, a zero-crossing method is used to determine the precise edge placements. The LoG filter is used in the Marr-Hildreth edge detector, which can recognise edges of different widths and orientations but is computationally expensive. Medical imaging applications like tumor and lesion detection frequently use the Marr-Hildreth edge detector.
The Marr-Hildreth edge detector consists of several steps:

- Smoothing
- Laplacian of Gaussian
- Zero-Crossing
- Edge Tracking

The Marr-Hildreth edge detector produces a binary picture in which the bright pixels correspond to the edges of the original image. The method is very adjustable, with parameters that may be tweaked to obtain various levels of edge detection sensitivity and accuracy. The Marr-Hildreth edge detector is a strong edge detection filter that can detect edges of various lengths and orientations, making it helpful in a variety of applications.

IV. CANNY FILTER

The quantity of data that has to be processed is greatly decreased using a technique called canny edge detection for eliminating important structural information from a variety of visual objects. It is widely used in a variety of computer vision systems. Canny discovered that the conditions for using edge detection in various vision systems are quite similar. Therefore, a system for edge

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detection that meets these requirements can be used in various contexts. These typical edge detection recommendations include:

Since edge detection has a low error rate, it should successfully identify most of the edges visible in the image.

The midpoint of the edge should be accurately localized as the operator's edge point.

When possible, image noise should be reduced, and an image edge should only be marked once.

Canny applied the calculus of variations, a technique for determining the function that maximizes a given functional, to meet these objectives. Although its first derivative of a Gaussian can be used as a close approximation, the optimal function of Canny's detector is represented by the sum of four exponential terms.

The Canny edge detection algorithm is one of the most precisely planned edge detection methods that has been created to date. It offers good and accurate detection. It swiftly ascended to become one of the most extensively used edge detection algorithms due to its capacity to meet all three requirements for edge detection as well as its ease of usage.

A. Edge detection process

The Canny edge detection algorithm's process may be divided into the following five steps:

- Use a Gaussian filter to smooth the image and remove noise.
- Look for the image's intensity gradients.
- Reduce spurious edge detection responses by using gradient magnitude thresholding or lower bound cut-off suppression.
- Consider using a twofold threshold to discover probable edges.
- Suppressing all further weak edges that are unrelated to strong edges brings edge identification for the hysteresis track to a conclusion

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B. Edge detection parameters

A variety of adjustable settings can alter the Canny algorithm calculating time and efficacy.

The magnitude of the Gaussian filter, which was employed as a smoothing filter in the first step, has a direct impact on how the Canny algorithm performs. Smaller filters provide less blur, making it easier to perceive thin, crisp lines. broader filters, which spread pixel values across a broader region of the image, provide more blur. Greater blurring radii are preferred for identifying larger, smoother edges, like the border of a rainbow.

Thresholds: Using two thresholds with hysteresis allows more versatility than a single-threshold strategy. A threshold that is set too high might result in the omission of crucial data. On the other side, if the threshold is set too low, noise could be misinterpreted for valuable, crucial information. It is difficult to provide a criterion that is applicable to all photographs. Currently, there isn't a tried-and-true solution to this problem.

C. Gaussian Filter

It is essential to filter out the noise to prevent false detection caused by it since noise in the picture may quickly invalidate any edge detection findings. A Gaussian filter kernel is convolutioned with the picture to smooth it out. This will considerably smooth the image and lessen the effect of visual noise on the edge detector. The following equation describes a kernel of size (2k+1)(2k+1) for a Gaussian filter:

$$H_{ij} = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(i-(k+1))^2 + (j-(k+1))^2}{2\sigma^2}\right); 1 \le i, j \le (2k+1)$$

….(1)

It is crucial to understand that the size of the Gaussian kernel employed will affect the detector's performance. The detector's size grows in proportion to its noise sensitivity. Additionally, as the size of the Gaussian filter kernel increases, so does the localization error for edge detection. A 5x5 is generally considered to be an acceptable size, however this may depend on the situation.

D. The intensity gradient of the image

The Canny approach employs four filters to differentiate between the horizontal, vertical, and diagonal edges in the blurred picture since an edge in an image might point in a number of different directions. An edge detection operator, such as Roberts, Prewitt, or Sobel, returns the horizontal (Gx) and vertical (Gy) components of the first derivative. One may deduce the edge gradient and direction from this:

$$\mathbf{G} = \sqrt{\mathbf{G}_x{}^2 + \mathbf{G}_y{}^2}$$

…….(2)

E. Log Double threshold

After non-maximum suppression, the edge pixels in a picture still present a more accurate depiction of actual edges. However, some edge pixels remain due to noise and color variation. To account for these deceptive reactions,

edge pixels with a low gradient value must be filtered out, whereas edge pixels with a high gradient value must be kept. This is done by choosing high and low threshold values. Strong edges are said to exist in each pixel with a gradient value higher than the high threshold value. The gradient value of a pixel with a weak edge is more than the low threshold value but less than the high threshold value. If a pixel's gradient value is below the low threshold value, its edge will be suppressed. Based on the content of the input image, two experimentally established threshold values are defined.

F. Edge tracking by hysteresis

The final edge picture should unquestionably contain the strong edge pixels, which are thought to originate from

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genuine edges in the image. However, it will be challenging to tell whether the weak edge pixels are brought on by noise, color variations, or an actual edge. Weak edge pixels shouldn't be considered in this scenario. This approach is based on the notion that noise responses are (often) connected to strong edge pixels but unrelated to weak edge pixels from genuine edges. Blob analysis is used to track the edge connection by looking at a weak edge pixel and its eight linked neighbouring pixels. As long as the blob contains at least one strong edge pixel, it may be determined that it has a weak edge point that has to be safeguarded. These weak edge pixels develop into strong edges, which preserves the weak edge pixels close by.

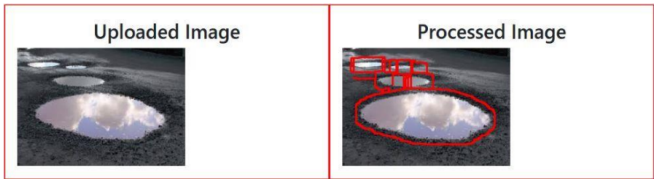


Fig 1 Canny edge detection applied to a photograph

F.1 Walkthrough of the algorithm

This section will show the progression of an image through each of the five steps as shown in Fig 2.

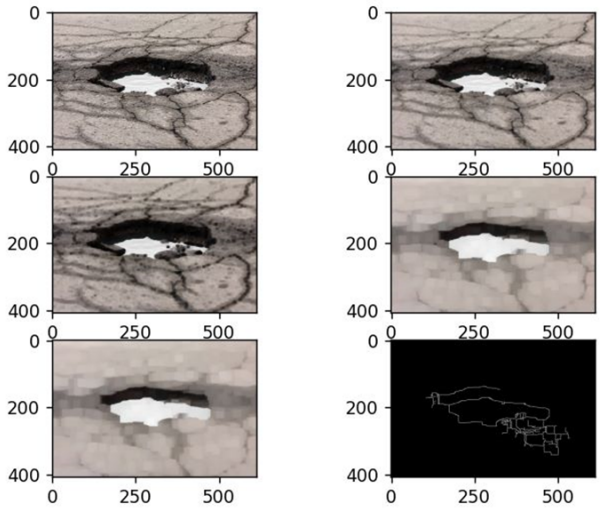


Fig 2 step by step edge detection

Picture segmentation comes after picture pre-processing as a crucial stage in the pothole detecting process. Different image segmentation techniques are used to separate the original pothole image from the road area that doesn't have potholes.

Edge detection technique

- Either a picture is taken and then filtered, or a filter immediately analyses the video input.
- Since a picture is often made up of a very limited number of colours or pixels, edge detection may take advantage of this property.
- As previously said, edges occur when the adjoining surface appears to be different. For example, in a pothole, the damaged region is often a little darker than the surrounding area, making it stand out from the rest of the image.
- This distinction is used in the creation of an image's edge.
- Now, the edge can even be caught by anything that is not similar to a road. This problem may be solved by employing the range of color gradient, or by specifying a certain range of darkness that separates the pothole.
- Thus, efficiency can be increased because it can only mark the edge if a pixel of that color is visible**.**

V. DATA LOGGING

Logging mechanism for data:

Data logging involves keeping track of the data or information that is obtained during an analysis by a device. It enables the storage or separation of a certain type of data in order to use it later for operations.

Data logging in this project may be done in two different methods.

1 - Localized data recording

The first option is using a gadget to visually survey the area, find potholes, and save data in a database at the same time is one technique to carry out the job. The primary elements being recorded here are the following: the area of potholes; the height/depth of potholes; the severity determined from this data; and the live position of the pothole. Location is made up of latitude and longitude, thus it has to be saved in order to monitor potholes precisely and take the appropriate steps.

2- Wireless or web-based data recording

The second option is for a third party to assist in finding the pothole using a picture and software, which is often provided with an application (mobile app). The other method is when a third party uses the software and image that are often provided in an application form to aid find the pothole. Using this method of operation, the location can be determined by using the meta data that is stored in an image when it is taken, and the rest of the process is the same.

This method of application could be less effective because it relies entirely on the efforts of the individual and his decision to participate in the process. Thus, the first method is faster, more reliable, and more accurate. There are several uses for this data. It may be used to assist in gathering information on potholes for authorities, It may also be combined with live maps to assist users in selecting the best path with the fewest obstacles.

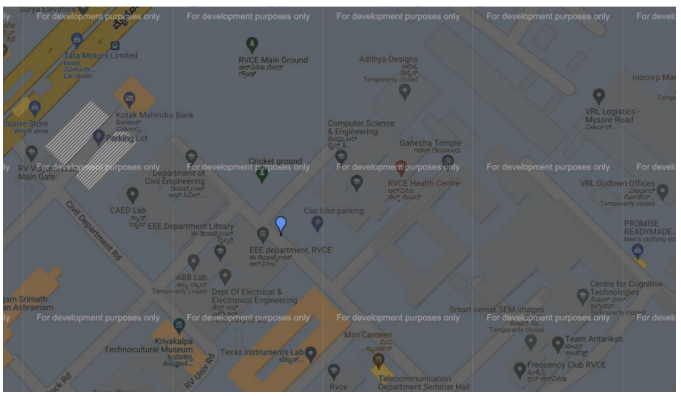


Fig 3 location of pothole

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VI. Conclusion

The project of detecting potholes using the Canny filter and data logging has been successful in achieving its objective. The Canny filter has been used effectively to detect the edges of the potholes in the images, and the data logging has allowed for the capture and storage of important information about the detected potholes.

This project has the potential to make a significant impact in the field of road maintenance and safety. By detecting potholes in real-time, authorities take quick action to repair them, preventing accidents and reducing the risk of damage to vehicles.

Furthermore, the use of data logging enables the collection of important information, such as the location and size of the detected potholes, which can be used for future analysis and planning.

Overall, this project demonstrates the power of image processing and data logging in addressing real-world problems and improving the safety and efficiency of infrastructure.

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